

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME:** Database Management Systems  
**COURSE CODE:** CSA05

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### COURSE OUTCOME COVERED

**CO3:** *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)  
Assessment Tool Weightage: 30%

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| QUESTIONS |                                                                                                                                                                                                           |                  | Total Marks: 50 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| Q.No      | Question                                                                                                                                                                                                  | Bloom's Level    | Marks           |
| 1         | Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies                 | BL2 – Understand | 5               |
| 2         | A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies | BL2 – Understand | 5               |
| 3         | Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF                                                                                                                  | BL2 – Understand | 5               |
| 4         | BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF                                                                                                                                            | BL2 – Understand | 5               |
| 5         | Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other                                                              | BL2 – Understand | 5               |
| 6         | Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly                                         | BL2 – Understand | 5               |
| 7         | A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An                                                           | BL2 – Understand | 5               |

|           |                                                                                                                                                                                             |                  |          |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------|
|           | attribute that depends on a non-prime attribute (d) A foreign key reference                                                                                                                 |                  |          |
| <b>8</b>  | If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$ , then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined | BL2 – Understand | <b>5</b> |
| <b>9</b>  | 5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form                                                            | BL2 – Understand | <b>5</b> |
| <b>10</b> | Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist                                | BL2 – Understand | <b>5</b> |

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**Code of Conduct:**

I CH. Durga Prasad (192425305) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

CH. Durga Prasad

**1. Which of the following is true about 1NF?**

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic**
- (d) Eliminates transitive dependencies

ANS: (c) All attributes must be atomic

**2. A relation is in 2NF if it is in 1NF and:**

- (a) Has no transitive dependencies
- (b) Has no partial dependencies on primary key**
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

ANS: (b) Has no partial dependencies on primary key

**3. Which normal form deals with multi-valued dependencies?**

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF**

ANS: (d) 4NF

**4. BCNF is a stricter version of:**

- (a) 1NF
- (b) 2NF
- (c) 3NF**
- (d) 4NF

ANS: (c) 3NF

**5. Functional dependency  $X \rightarrow Y$  means:**

- (a) Y determines X
- (b) X determines Y
- (c) X and Y are unrelated
- (d) Both determine each other

ANS: (b) X determines Y

**6. Which of the following anomaly is NOT associated with unnormalized relations?**

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly**

ANS: (d) Retrieval anomaly

**7. A superkey is:**

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples**
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

ANS: (b) Any set of attributes that uniquely identifies tuples

**8. If  $R(A,B,C)$  has FD:  $A \rightarrow B$  and  $B \rightarrow C$ , then C is:**

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A**
- (d) Independently determined

ANS; (c) Transitively dependent on A

**9. 5NF is also known as:**

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form**
- (d) Element Normal Form

ANS: (c) Project-Join Normal Form

**10. Lossless-join decomposition ensures:**

- (a) No data is lost during decomposition**
- (b) All FDs are preserved
- (c) The relation is in BCNF
- (d) No null values exist

ANS: (a) No data is lost during decomposition